Pearson Edexcel International A Level · Pure Mathematics 1

Chapter 6 — Trigonometric ratios
Original Practice Questions

30 complete questions · Questions followed by answers and suggested mark schemes

- Answer all parts and show sufficient working.
- Give exact answers unless a degree of accuracy is requested.
- Angles are in radians unless degrees are explicitly indicated.
- Diagrams are not necessarily drawn to scale.
- The marks for each subpart are shown in brackets.

P1 第6章 · 三角比
英文题目 · 中文解析 · 建议评分细则
本册30题；P1、P2、P3全套25章共750题。

内部复核稿，未完成外部教师逐题审阅。
Original practice material. Not a Pearson examination paper.

Triangle ABC has $AB = 9$ cm, $AC = 7$ cm and $\angle BAC = 60°$.

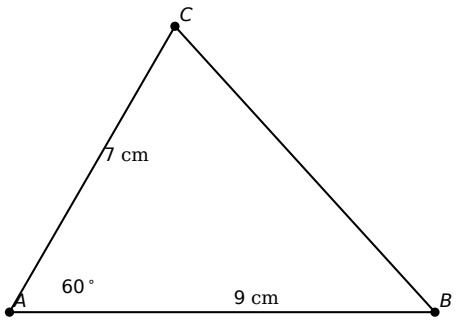


Figure 1

(a) Find the exact length of BC. (3)

(b) Find the exact area of the triangle. (2)

(Total for Question 1 is 5 marks)

2. P1-C06-R02 · P1 · Chapter 6

Triangle PQR has $PQ = 12$ m, $\angle PQR = 41^\circ$ and $\angle PRQ = 67^\circ$.

(a) Find $\angle QPR$. (2)

(b) Giving lengths to 3 significant figures, find

(i) PR, (2)

(ii) the perimeter of the triangle. (3)

(Total for Question 2 is 7 marks)

3.

In triangle ABC, $AB = 10$ cm, $BC = 8$ cm and $\angle BAC = 35°$.

(a) Use the sine rule to show that two values of $\angle ACB$ are possible. Give both values to 1 decimal place. (3)

(b) For the triangle in which $\angle ACB$ is obtuse, find AC to 3 significant figures. (3)

(Total for Question 3 is 6 marks)

4.

A triangular panel has sides 6 cm, 9 cm and 11 cm.

(a) Find its largest angle to 1 decimal place. (3)

(b) Find its area to 3 significant figures. (2)

(Total for Question 4 is 5 marks)

5.

Two paths leave a station S. A point A lies 120 m due east of S, and a point B lies 150 m from S on a bearing of 040°.

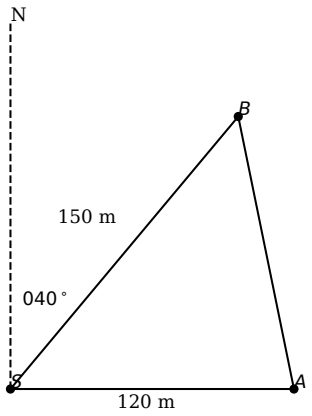


Figure 5

(a) Find $\angle ASB$. (2)

(b) Find AB and the area of triangle SAB, both to 3 significant figures. (4)

(Total for Question 5 is 6 marks)

A vertical tower stands on horizontal ground. From point A, the angle of elevation to its top is $28°$. Point B is 30 m closer to the tower on the same straight line, and its angle of elevation is $43°$.

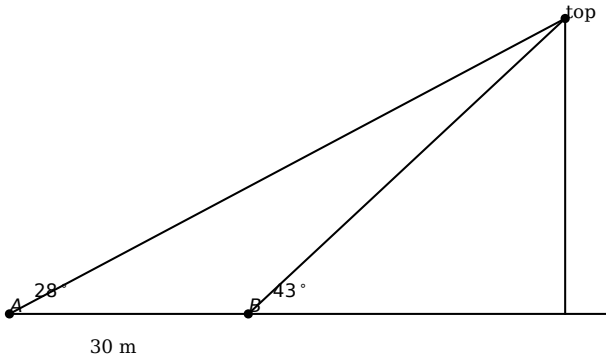


Figure 6

(a) Form an equation in d, the horizontal distance from B to the foot of the tower. (2)

(b) Find the height of the tower to 3 significant figures. (3)

(Total for Question 6 is 5 marks)

7.

In a triangle, two sides have lengths 5 cm and 8 cm and the included angle is θ. Its area is $10\sqrt{3}\ \text{cm}^2$.

(a) Find all possible values of θ in $0^\circ < \theta < 180^\circ$. (3)

(b) Given that the third side is longer than 8 cm, find its exact length. (3)

(Total for Question 7 is 6 marks)

8.

The graph $y = 3\sin x$ is considered for $0^\circ \leq x \leq 360^\circ$.

(a) Sketch the graph, marking all intercepts and all maximum and minimum points. (3)

(b) Find the values of x for which $y = \frac{3}{2}$. (2)

(Total for Question 8 is 5 marks)

A triangular support has sides $AB = AC = 13$ cm and base $BC = 10$ cm.

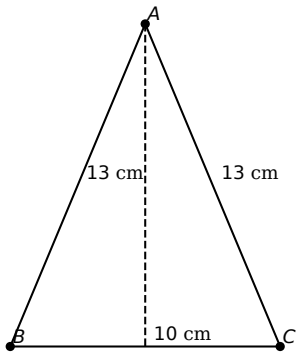


Figure 9

(a) Find the perpendicular height from A to BC and hence the area. (3)

(b) Find $\angle BAC$ to 1 decimal place. (3)

(Total for Question 9 is 6 marks)

Point B is 80 m due north of A. Point C is on a bearing of 065° from A and a bearing of 120° from B.

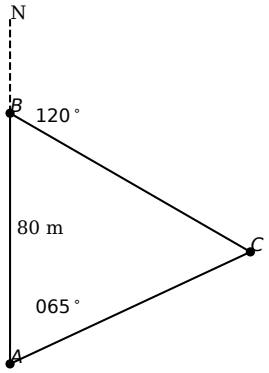


Figure 10

(a) Find all three angles in triangle ABC. (3)

(b) Find AC to 3 significant figures. (2)

(Total for Question 10 is 5 marks)

The curve is $y = 2\cos x - 1$, for $0^\circ \leq x \leq 360^\circ$.

(a) State the maximum and minimum values of y, and all values of x at which they occur. (3)

(b) Find the coordinates where the curve crosses the x-axis. (2)

(Total for Question 11 is 5 marks)

In triangle ABC, $AB = 15$ cm, $AC = 10$ cm and $\angle ABC = 30°$.

(a) Find the two possible values of $\angle ACB$. (3)

(b) For the triangle in which $\angle ACB$ is acute, find

(i) $\angle BAC$ to 1 decimal place, (2)

(ii) the area to 3 significant figures. (2)

(Total for Question 12 is 7 marks)

A ladder of length 7 m leans against a vertical wall on horizontal ground. Its angle with the ground is 64°.

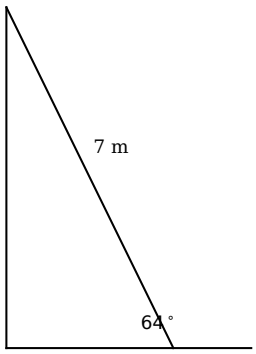


Figure 13

(a) Find the height reached on the wall to 3 significant figures. (2)

(b) The top is lowered by 1 m while the ladder remains against the wall. Find the new angle with the ground to 1 decimal place. (3)

(Total for Question 13 is 5 marks)

A parallelogram has adjacent sides 9 cm and 14 cm and an included acute angle of 38°.

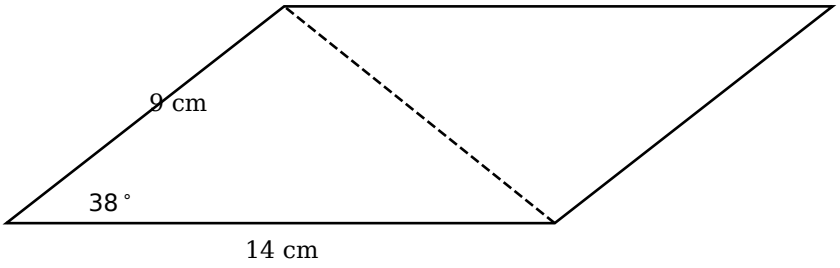


Figure 14

(a) Find its area to 3 significant figures. (2)

(b) Find the length of its shorter diagonal to 3 significant figures. (3)

(Total for Question 14 is 5 marks)

15.

Triangle ABC has $AB = 4$ cm, $AC = 7$ cm and $BC = \sqrt{37}$ cm.

(a) Find the exact value of $\cos A$ and hence find A. (3)

(b) Find the exact area of the triangle. (2)

(Total for Question 15 is 5 marks)

16.

For $0^\circ < x < 180^\circ$, $\sin x = \frac{5}{13}$ and x is obtuse.

(a) Find the exact value of $\cos x$. (2)

(b) Find the exact value of $\tan x$. (2)

(Total for Question 16 is 4 marks)

17.

The graph $y = \tan x$ is considered for $-90^\circ < x < 270^\circ$.

(a) Sketch the graph, showing all vertical asymptotes and axis intersections in this interval. (3)

(b) Find all x in the stated interval for which $y = -1$. (2)

(Total for Question 17 is 5 marks)

A triangular field has sides 120 m and 160 m enclosing an angle of $47°$. A straight fence joins the free ends of those sides.

(a) Find the length of the fence to the nearest metre. (3)

(b) A path runs along the 160 m side and has perpendicular width 2 m inside the field. For this calculation only, its area is modelled as a rectangle of dimensions 160 m by 2 m. Find

(i) the modelled path area, (1)

(ii) the percentage of the field area occupied by this modelled path, to 1 decimal place. (3)

(Total for Question 18 is 7 marks)

Two ships leave a harbour at the same time. After 2 hours, ship A has travelled 30 km on a bearing of 020° and ship B has travelled 40 km on a bearing of 110°.

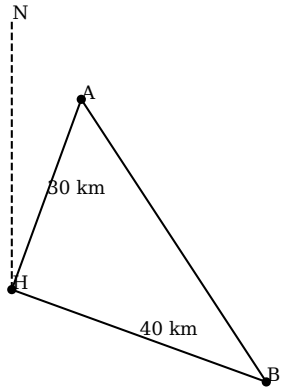


Figure 19

(a) Find their separation at this time. (3)

(b) Find the angle between the direction from A to the harbour and the direction from A to B, to 1 decimal place. (3)

(Total for Question 19 is 6 marks)

A triangle has sides x cm, $2x$ cm and $\sqrt{19}$ cm. The angle between the sides of lengths x and $2x$ is 120°, and $x > 0$.

(a) Find the exact value of x. (3)

(b) Find the exact area of the triangle. (2)

(Total for Question 20 is 5 marks)

21. P1-C06-R21 · P1 · Chapter 6

The curve is $y = 4\sin(2x)$, where angles are in degrees and $0^\circ \leq x \leq 180^\circ$.

(a) State its period and the coordinates of its maximum and minimum points on this interval. (3)

(b) Find all zeros on the interval. (2)

(Total for Question 21 is 5 marks)

Triangle ABC has area 24 cm^2, $AB = 8$ cm and $AC = 10$ cm.

(a) Find the two possible values of $\angle BAC$ to 1 decimal place. (3)

(b) Find both possible exact lengths of BC. (3)

(Total for Question 22 is 6 marks)

A mast is held by two wires attached to the same top point. The wires are fixed to the horizontal ground at opposite sides of the mast, at distances 6 m and 10 m from its foot. The shorter wire makes angle 58° with the ground.

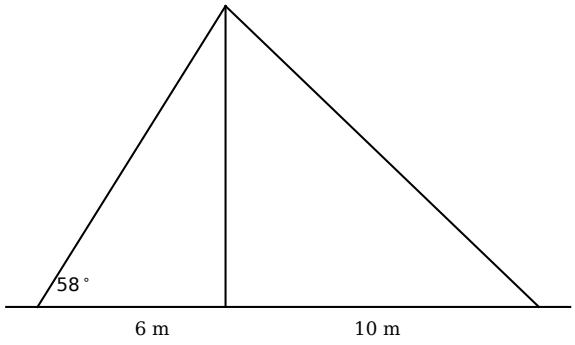


Figure 23

(a) Find the mast height to 3 significant figures. (2)

(b) For the longer wire, find

(i) its angle with the ground to 1 decimal place, (2)

(ii) its length to 3 significant figures. (2)

(Total for Question 23 is 6 marks)

For $0^\circ \leq x \leq 360^\circ$, $\cos x = -\frac{1}{2}$.

(a) Find all possible values of x. (2)

(b) For each value of x found in part (a), state

(i) the exact value of $\sin x$, (2)

(ii) the exact value of $\tan x$. (2)

(Total for Question 24 is 6 marks)

A triangle has sides 7 cm, 8 cm and 9 cm. Let θ be the angle opposite the 8 cm side.

(a) Show that $\cos\theta = \frac{11}{21}$. (2)

(b) Hence find the exact area of the triangle. (3)

(Total for Question 25 is 5 marks)

26.

The height of a point on a wheel is modelled by $h = 5 - 4\cos\theta$ metres, for $0^\circ \leq \theta \leq 360^\circ$.

(a) Find the minimum and maximum heights and state the angle at which each occurs. (3)

(b) Find all angles at which the height is 7 m. (2)

(Total for Question 26 is 5 marks)

$ABCD$ is a quadrilateral with diagonal $AC = 12$ cm. In triangle ABC, $AB = 7$ cm and $BC = 9$ cm. In triangle ACD, $AD = CD = 10$ cm. Points B and D lie on opposite sides of AC.

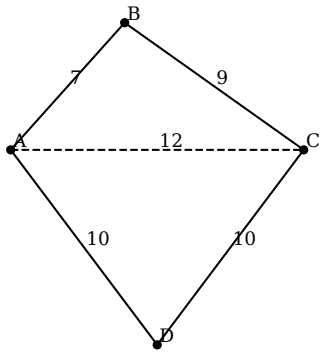


Figure 27

(a) Find $\angle ABC$ to 1 decimal place. (2)

(b) Find the area of the quadrilateral to 3 significant figures. (4)

(Total for Question 27 is 6 marks)

A hillside rises at 18° to the horizontal. A straight path on the hillside is 90 m long.

(a) Find the vertical rise and horizontal distance, each to 3 significant figures. (3)

(b) An alternative route to a platform at the same height consists of a horizontal section of length 100 m followed by a vertical climb. Find its total length to 3 significant figures. (2)

(Total for Question 28 is 5 marks)

In triangle ABC, $AB = 6$ cm and $AC = 10$ cm. Point D lies on AC with $AD = 4$ cm. It is given that $\angle BAC = 60°$.

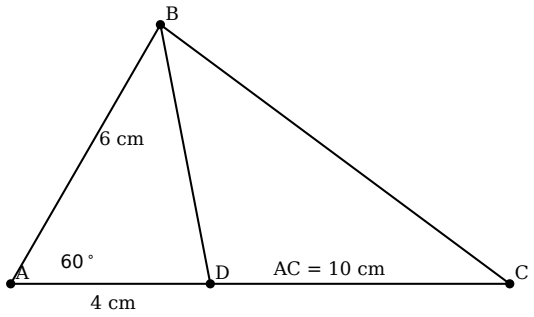


Figure 29

(a) Find the exact length of BD. (2)

(b) For the same diagram, find

(i) the exact area of triangle BCD, (3)

(ii) the ratio of the areas of ABD and BCD. (2)

(Total for Question 29 is 7 marks)

The curve C is $y = 1 + 2\cos(3x)$, where angles are in degrees and $0^\circ \leq x \leq 120^\circ$.

(a) State the period and the coordinates of all maximum and minimum points on this interval. (3)

(b) Find all x in the interval for which $y = 0$. (3)

(Total for Question 30 is 6 marks)

Answers and Suggested Mark Schemes

中文解答与建议评分点

M：方法分；A：准确分，通常依赖对应方法；B：独立得分点。
等价正确解法应获相应分数。评分点为原创建议，并非官方评分细则。

Question 1

P1-C06-R01 · 5 marks

(a) · 3 marks

$\sqrt{67}$ cm.

余弦定理$BC^2=81+49-2\times9\times7\times1/2=67$。

M1	正确余弦定理
M1	使用60度精确余弦
A1	最简根式

(b) · 2 marks

$\frac{63\sqrt{3}}{4}$ cm^2.

面积$1/2\times9\times7\times\sin60°$，代入$\sqrt{3}/2$。

M1	两边夹角面积式
A1	精确面积

Question 2

P1-C06-R02 · 7 marks

(a) · 2 marks

72°.

三角形内角和180°。

M1	内角和
A1	72度

(b) Giving lengths to 3 significant figures, find

(i) · 2 marks

$12\sin 41^\circ / \sin 67^\circ \approx 8.55$ m.

已知边PQ对67°，PR对41°，正弦定理配对应边角。

M1	正弦定理对应
A1	精度及单位

(ii) · 3 marks

$12 + 12(\sin 41^\circ + \sin 72^\circ)/\sin 67^\circ \approx 33.0$ m.

另边QR对72°，用未舍入的三边相加。

M1	正确求第三边
M1	用未舍入值求周长
A1	周长

Question 3

P1-C06-R03 · 6 marks

(a) · 3 marks

45.8° and 134.2°.

sinC=10sin35°/8≈0.7169705，两角为arcsin及180°−arcsin，且与35°相加均小于180°。

M1	正确正弦关系
M1	找补角
A1	两角及合法检查

(b) · 3 marks

$AC \approx 2.61$ cm.

用未舍入的钝角C得B=180−35−C，再由AC=8sinB/sin35°。

M1	选钝角并求余角
M1	再用正弦定理
A1	长度

Question 4

P1-C06-R04 · 5 marks

(a) · 3 marks

92.1°.

最大角对11；$\cos\theta=(36+81-121)/(2\times6\times9)=-1/27$。

M1	对应最长边与最大角
M1	余弦定理
A1	角度

(b) · 2 marks

$27.0\ \text{cm}^2$.

面积$(1/2)\times6\times9\times\sin\theta$，使用未舍入$\theta$。

M1	正确两边夹角面积
A1	精度及单位

Question 5

P1-C06-R05 · 6 marks

(a) · 2 marks

50°.

正东方位角90°，与040°的夹角50°。

M1	从北向顺时针理解方位角
A1	夹角

(b) · 4 marks

$AB \approx 117$ m; area ≈ 6890 m^2.

用余弦定理求两端距离；面积$1/2 \times 120 \times 150 \times \sin 50^\circ$。

M1	余弦定理
A1	距离
M1	面积式
A1	面积

Question 6

P1-C06-R06 · 5 marks

(a) · 2 marks

$d\tan 43^\circ = (d+30)\tan 28^\circ$.

同一塔高分别为d tan43°与(d+30)tan28°；测量点与塔脚共线，地面水平。

M1	写两种塔高
A1	距离差使用正确

(b) · 3 marks

$h = 30\tan 28^\circ \tan 43^\circ / (\tan 43^\circ - \tan 28^\circ) \approx 37.1$ m.

先解d再乘tan43°，保留中间精度。

M1	解水平距离
M1	求塔高
A1	精度与单位

Question 7

P1-C06-R07 · 6 marks

(a) · 3 marks

$60^\circ, 120^\circ$.

面积$20\sin\theta = 10\sqrt{3}$，所以$\sin\theta = \sqrt{3}/2$，在三角形范围有两个角。

M1	面积方程
M1	两个正弦解
A1	完整角度

(b) · 3 marks

$\sqrt{129}$ cm.

$\theta = 60^\circ$时第三边$^2 = 49$，$\theta = 120^\circ$时为129；长度大于8只允许后者。

M1	分别检验两夹角或比较余弦
M1	用长度条件选支
A1	根式长度

Question 8

P1-C06-R08 · 5 marks

(a) · 3 marks

Intercepts $(0,0),(180,0),(360,0)$; maximum $(90,3)$; minimum $(270,-3)$.

正弦一个完整周期，幅值3，顺序从零上升。

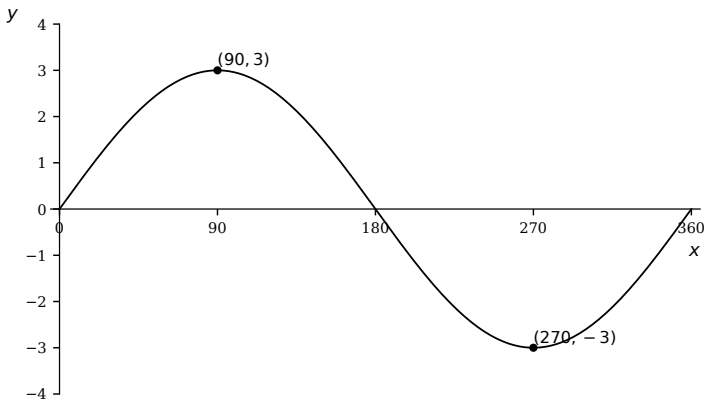
B1	三个零点
B1	两极值
A1	正确正弦形状

(b) · 2 marks

$x = 30^\circ, 150^\circ$.

sinx=1/2，第一和第二象限各一解。

M1	正确化简方程
A1	全部解



Suggested sketch

Question 9

P1-C06-R09 · 6 marks

(a) · 3 marks

Height 12 cm; area 60 cm^2.

等腰三角形高平分底边，半底5，勾股 $\sqrt{(169-25)}=12$。

M1	平分底边并勾股
A1	高
A1	面积

(b) · 3 marks

45.2°.

半顶角满足$\sin(\theta/2)=5/13$，求arcsin再乘2。

M1	正确半角直角三角关系
M1	恢复完整顶角
A1	角度

Question 10

P1-C06-R10 · 5 marks

(a) · 3 marks

$A = 65^\circ$, $B = 60^\circ$, $C = 55^\circ$.

BA从B看向正南180°，与BC的120°夹60°；最后由角和求55°。

M1	正确两处方位夹角
A1	两已知角
A1	第三角

(b) · 2 marks

$80 \sin 60^\circ / \sin 55^\circ \approx 84.6$ m.

AC对B的60°，AB对C的55°，用正弦定理。

M1	正确对应边角
A1	长度

Question 11

P1-C06-R11 · 5 marks

(a) · 3 marks

Maximum 1 at $0^\circ, 360^\circ$; minimum -3 at 180°.

cos取1时最大，取−1时最小，注意区间包含两个端点。

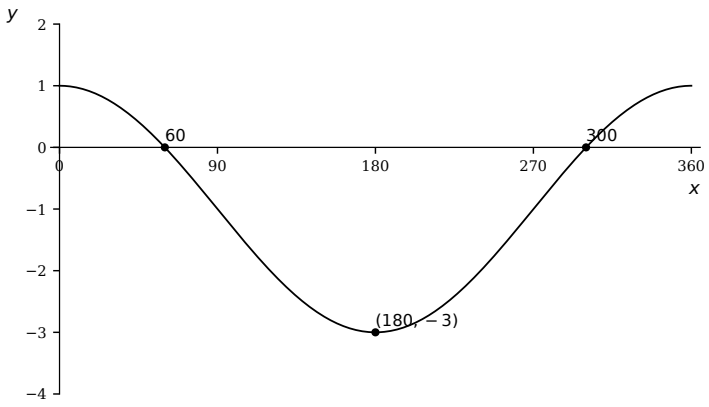
M1	用余弦范围
A1	两极值
A1	所有位置

(b) · 2 marks

$(60, 0)$ and $(300, 0)$, with x in degrees.

令cosx=1/2，第一与第四象限两个解。

M1	化简方程
A1	两个交点



Suggested sketch

Question 12

P1-C06-R12 · 7 marks

(a) · 3 marks

$\sin C = \frac{3}{4}$, so $C \approx 48.6^\circ$ or 131.4°.

正弦定理sinC/15=sin30°/10；两角均与B组成合法三角形。

M1	正弦关系
M1	补角支
A1	两角

(b) For the triangle in which $\angle ACB$ is acute, find

(i) · 2 marks

101.4°.

A=180−30−arcsin(3/4)，用未舍入角。

M1	选正确C并用内角和
A1	A角

(ii) · 2 marks

$\approx 73.5\ \text{cm}^2$.

面积1/2×15×10×sinA。

M1	正确面积式
A1	面积

Question 13

P1-C06-R13 · 5 marks

(a) · 2 marks

$7\sin 64^\circ \approx 6.29$ m.

高度是该角对边，梯长是斜边。

M1	正弦关系
A1	高度

(b) · 3 marks

$\arcsin\big((7\sin 64^\circ - 1)/7\big) \approx 49.1^\circ$.

先从原高度减1，再与不变梯长7构成正弦比；取0°到90°的角。

M1	正确新高度
M1	反求锐角
A1	角度

Question 14

P1-C06-R14 · 5 marks

(a) · 2 marks

$126\sin 38^\circ \approx 77.6\ \text{cm}^2$.

以一边为底、另一边的垂直分量为高；平行四边形面积不是三角形的一半。

M1	正确底高关系
A1	面积

(b) · 3 marks

$\sqrt{9^2+14^2-2(9)(14)\cos 38^\circ} \approx 8.86\ \text{cm}$.

较短对角线对锐角38°，使用余弦定理；另一条对补角更长。

M1	选择锐角对应短对角线
M1	余弦定理
A1	长度

Question 15

P1-C06-R15 · 5 marks

(a) · 3 marks

$\cos A = \frac{1}{2}$, $A = 60^\circ$.

余弦定理(16+49−37)/(2×4×7)=28/56。

M1	正确余弦关系
A1	精确余弦
A1	角度

(b) · 2 marks

$7\sqrt{3}\ \text{cm}^2$.

面积1/2×4×7×√3/2。

M1	正确面积式
A1	根式面积

Question 16

P1-C06-R16 · 4 marks

(a) · 2 marks

$-\frac{12}{13}$.

参考直角三角形对边5斜边13，邻边12；钝角的余弦为负。

M1	勾股得到参考边12
A1	正确负号与比值

(b) · 2 marks

$-\frac{5}{12}$.

tan为sin/cos，正除负得负。

M1	正确三角比商
A1	最简值

Question 17

P1-C06-R17 · 5 marks

(a) · 3 marks

Asymptotes $x = -90^\circ, 90^\circ, 270^\circ$ at boundaries or inside; zeros $0^\circ, 180^\circ$.

正切周期180°，奇数90°为竖直渐近线，各分支严格上升；边界虽不取值仍标渐近线。

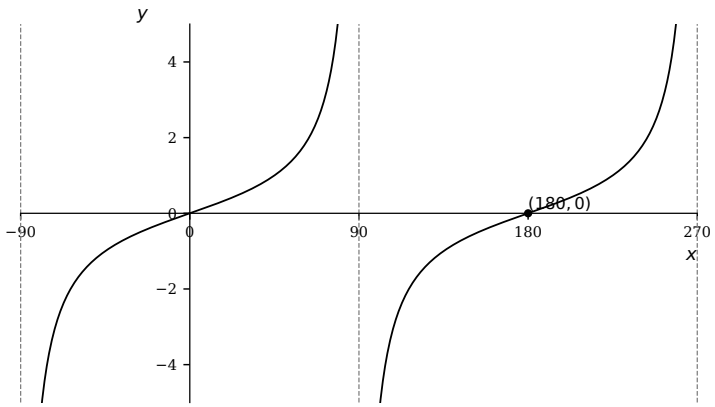
B1	渐近线
B1	两零点
A1	正确分支走向

(b) · 2 marks

$x = -45^\circ, 135^\circ$.

tan为−1的参考角45°，每180°重复，筛选开区间。

M1	使用周期并筛选
A1	全部解



Suggested sketch

Question 18

P1-C06-R18 · 7 marks

(a) · 3 marks

118 m.

余弦定理给 $\sqrt{(120^2+160^2-2\times120\times160\cos47^\circ)}$，再按米四舍五入。

M1	正确余弦式
M1	计算未舍入长度
A1	最近整数米

(b) A path runs along the 160 m side and has perpendicular width 2 m inside the field. For this calculation only, its area is modelled as a rectangle of dimensions 160 m by 2 m. Find

(i) · 1 marks

$320\ \text{m}^2$.

按明确给出的矩形近似模型计算。

B1	矩形面积

(ii) · 3 marks

$100(320)/(9600\sin47^\circ)\approx4.6\%$.

全场面积为$9600\sin47^\circ$；用近似路径面积除以全场面积再乘100。

M1	正确三角场地面积
M1	百分比关系
A1	百分比

Question 19

P1-C06-R19 · 6 marks

(a) · 3 marks

50 km.

夹角110−20=90°，勾股 $\sqrt{(30^2+40^2)}=50$。

M1	从方位得直角
M1	勾股
A1	距离

(b) · 3 marks

53.1°.

在A处夹角的对边HB=40，斜边AB=50，所以sin角=4/5；取三角形锐角。

M1	正确识别A处角的边
M1	反求角
A1	精度

Question 20

P1-C06-R20 · 5 marks

(a) · 3 marks

$x = \sqrt{19/7}$.

余弦定理$19=x^2+4x^2-4x^2\cos120^\circ=7x^2$，取正根。

M1	正确含参数余弦式
M1	使用精确余弦
A1	正根

(b) · 2 marks

$\dfrac{19\sqrt{3}}{14}\ \text{cm}^2$.

面积$x^2\sin120^\circ=(19/7)(\sqrt{3}/2)$。

M1	正确面积式
A1	精确结果

Question 21

P1-C06-R21 · 5 marks

(a) · 3 marks

Period $180°$; maximum $(45, 4)$, minimum $(135, -4)$.

输入角2x走完360°时x走180°，故周期180°；极值在2x=90°,270°。

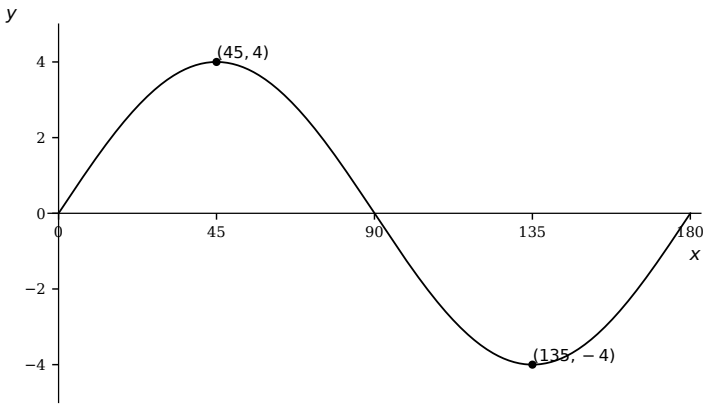
M1	正确处理周期
A1	极值位置
A1	极值坐标

(b) · 2 marks

$x = 0°, 90°, 180°$.

2x=0°,180°,360°，横坐标除2。

M1	输入角范围及零点
A1	全部x



Suggested sketch

Question 22

P1-C06-R22 · 6 marks

(a) · 3 marks

36.9° or 143.1°.

$40\sin A=24$，故$\sin A=3/5$；在0°到180°内有锐角和其补角。

M1	正确面积关系
M1	找两个角
A1	角度

(b) · 3 marks

6 cm and $2\sqrt{73}$ cm.

相应$\cos A=\pm 4/5$。余弦定理$BC^2=164-160\cos A$，分别36和292。

M1	找两余弦的正负
M1	余弦定理
A1	两精确长度

Question 23

P1-C06-R23 · 6 marks

(a) · 2 marks

$6\tan 58^\circ \approx 9.60$ m.

短线对应较近锚点6m；高度为6tan58°。

M1	正确识别短线锚点
A1	高度

(b) For the longer wire, find

(i) · 2 marks

$\arctan\left((6\tan 58^\circ)/10\right) \approx 43.8^\circ$.

同一塔高除以较远水平距离10构成正切比。

M1	正确正切关系
A1	角度

(ii) · 2 marks

$\sqrt{100 + 36\tan^2 58^\circ} \approx 13.9$ m.

水平直角边10与塔高用勾股。

M1	勾股求线长
A1	长度

Question 24

P1-C06-R24 · 6 marks

(a) · 2 marks

$120^\circ, 240^\circ$.

余弦负在第二和第三象限，参考角60°。

M1	正确参考角与象限
A1	全部解

(b) For each value of x found in part (a), state

(i) · 2 marks

$\sqrt{3}/2$ at 120°; $-\sqrt{3}/2$ at 240°.

参考角60°，正弦在第二象限正、第三负。

M1	正确参考值
A1	与两角配对

(ii) · 2 marks

$-\sqrt{3}$ at 120°; $\sqrt{3}$ at 240°.

用$\sin/\cos$或参考角符号规则。

M1	正确比值及符号
A1	两值

Question 25

P1-C06-R25 · 5 marks

(a) · 2 marks

$\cos\theta = (7^2 + 9^2 - 8^2)/(2\cdot 7\cdot 9) = 11/21.$

分子49+81−64=66，分母126，约为11/21。

M1	正确余弦定理
A1	精确化简

(b) · 3 marks

$12\sqrt{5}\ \text{cm}^2.$

θ是三角形内角且cos为正，所以sinθ=√(441−121)/21=8√5/21；面积(1/2)×7×9×该比。

M1	由正弦余弦关系或参考直角三角形得正弦
M1	正确面积式
A1	精确面积

Question 26

P1-C06-R26 · 5 marks

(a) · 3 marks

Minimum 1 m at $0^\circ, 360^\circ$; maximum 9 m at 180°.

余弦前系数负，cos=1时最低，cos=−1时最高；包含两端点。

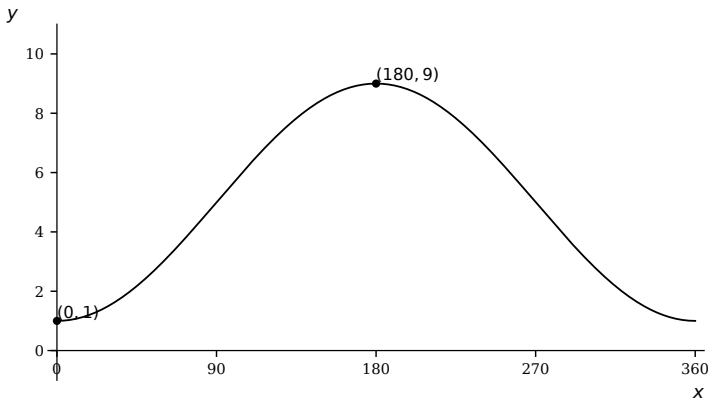
M1	正确使用负系数
A1	两高度
A1	所有角度

(b) · 2 marks

$120^\circ, 240^\circ$.

$5-4\cos\theta=7$给$\cos\theta=-1/2$。

M1	正确化简
A1	两角



Suggested sketch

Question 27

P1-C06-R27 · 6 marks

(a) · 2 marks

96.4°.

AC为角B对边，cosB=(49+81−144)/(2×7×9)=−1/9。

M1	正确余弦式
A1	角度

(b) · 4 marks

$\approx 79.3\ \text{cm}^2$.

ABC面积=(1/2)×7×9sinB=14√5≈31.305；ACD高√(100−36)=8，面积48；相加约79.3。

M1	ABC面积
M1	等腰分割求ACD面积
M1	两三角形相加
A1	总面积

Question 28

P1-C06-R28 · 5 marks

(a) · 3 marks

Rise 27.8 m; horizontal distance 85.6 m.

以路径为斜边，竖直分量$90\sin 18^\circ$，水平$90\cos 18^\circ$。

M1	两分量正确三角比
A1	竖直距离
A1	水平距离

(b) · 2 marks

128 m.

平台与原终点同高，所以竖直爬升仍为$90\sin 18^\circ$；两段总长$100+90\sin 18^\circ \approx 127.812$。

M1	给定段长相加
A1	长度

Question 29

P1-C06-R29 · 7 marks

(a) · 2 marks

$2\sqrt{7}$ cm.

在ABD中，$BD^2=36+16-2\times6\times4\times1/2=28$。

M1	在正确子三角形用余弦定理
A1	根式

(b) For the same diagram, find

(i) · 3 marks

$9\sqrt{3}$ cm^2.

DC=6，与AD共用对AC的高$6\sin60°=3\sqrt{3}$；面积$6\times3\sqrt{3}/2$。

M1	正确DC长度
M1	共同垂直高
A1	面积

(ii) · 2 marks

$2:3$.

同高三角形面积比等于底AD:DC=4:6。

M1	使用共同高
A1	最简比

Question 30

P1-C06-R30 · 6 marks

(a) · 3 marks

Period 120°; maxima $(0,3),(120,3)$; minimum $(60,-1)$.

输入角3x走过0°至360°；周期360/3=120，正系数余弦极值按标准位置缩放。

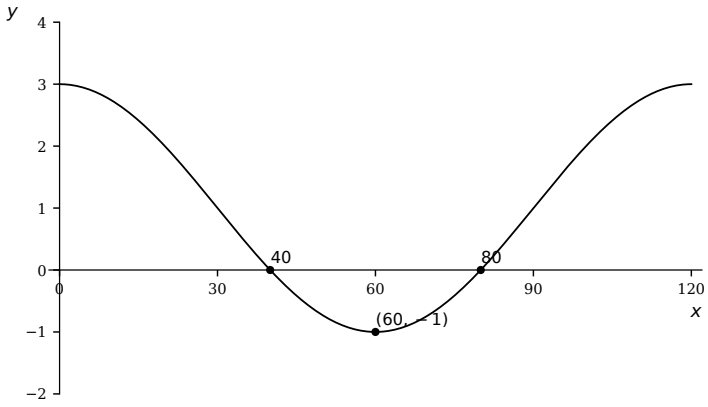
M1	周期缩放
A1	两最大点
A1	最小点

(b) · 3 marks

$x = 40^\circ, 80^\circ$.

cos3x=−1/2，3x=120°、240°；除3。

M1	解输入角方程
M1	换回x
A1	全部解



Suggested sketch